

Internet of Things (IoT)

Question Bank — Unit 1 & Unit 2

Unit 1: Introduction to IoT | Unit 2: Technology for IoT

UNIT 1: Introduction to IoT

Topics: Introduction to IoT, technology evolution, IoT architectures, resource management, communication protocols, standards and regulations.

Section A — Short Answer (2–3 Marks)

1. Define IoT. How does it differ from M2M (Machine-to-Machine) communication?
2. List any four characteristics of IoT systems.
3. What is meant by “resource-constrained” devices in the context of IoT? Give two examples.
4. Name any four communication protocols used at the network layer in IoT.
5. What is the role of a regulatory body (e.g., IEEE, ITU, IETF) in IoT standardisation?

Section B — Medium Answer (5–7 Marks)

6. Trace the technology evolution that led to the emergence of IoT (RFID → WSN → M2M → IoT). Highlight the key enabling shift at each stage.
7. Compare the three-layer and five-layer IoT reference architectures with a labelled diagram.
9. Differentiate between MQTT and CoAP as IoT application-layer protocols, in terms of transport, messaging model, and typical use case.
10. What is resource management in IoT? Discuss the challenges of managing compute, energy, and network resources on constrained devices.
11. Explain the role of 6LoWPAN in enabling IoT communication over low-power wireless networks.
12. Discuss why interoperability standards (e.g., oneM2M) are necessary for large-scale IoT deployment.

UNIT 2: Technology for IoT

Topics: Programming for IoT, different frameworks, virtualization concepts, embedded platforms.

Section A — Short Answer (2–3 Marks)

16. What is an embedded platform? Give two examples used for IoT prototyping.
17. Name any three programming frameworks/environments used for IoT development.
18. Define virtualization. How is it different from containerization?
19. What is an RTOS? Why is FreeRTOS preferred over a general-purpose OS for constrained IoT devices?
20. List two advantages of using MicroPython for rapid IoT prototyping.

Section B — Medium Answer (5–7 Marks)

21. Compare and contrast at least three IoT programming frameworks (e.g., Arduino, ESP-IDF, MicroPython, FreeRTOS-based bare-metal) in terms of abstraction level, real-time capability, memory footprint, and learning curve.

- 22.** Discuss the architecture of FreeRTOS (tasks, scheduler, queues, and semaphores) and explain how it supports real-time multitasking on a microcontroller such as the ESP32.
- 23.** Differentiate between a microcontroller-based embedded platform (e.g., ESP32/AVR) and a single-board computer (e.g., Raspberry Pi) for IoT applications.

Numerical

1. A sensor in an IoT network generates 200 bytes of data in 5 seconds. What is the data rate of this sensor in bits per second (bps)?
2. An IoT device consumes 10 mA of current when active and 0.5 mA in sleep mode. If the device is active for 2 hours and sleeps for 22 hours each day, and the battery capacity is 1000 mAh, what is the battery life in days?
3. An IoT sensor node sends data packets of size 512 bytes to a central server over a network with a bandwidth of 1 Mbps. Calculate the transmission latency for a single packet.
4. If a temperature sensor sends data every 10 seconds, and each data packet is 100 bytes, how much data will be stored in 1 day?
5. An IoT device operates at a frequency of 2.4 GHz with a transmission power of 10 dBm. If the signal loss is 100 dB, what is the received signal strength at the receiving end?
6. An IoT device transmits data at 5 Mbps, while the network bandwidth is 20 Mbps. What is the percentage of bandwidth utilization?
7. An IoT sensor has an average power consumption of 1.5 mW. If it operates for 12 hrs a day, how much energy does it consume in a month (30 days)?
8. A vibration sensor requires a sampling rate of 2000 Hz to capture the data accurately. How many samples will the sensor generate in 1 minute?
9. An IoT device sends 1000 packets, but only 920 packets are received successfully. What is the packet loss rate?
10. In a mesh network of 5 IoT nodes, each node sends 50 KB of data per hour to a central node. Calculate the total data transmitted to the central node in a day.